CS 185/285 Midterm (Practice Questions)

Student ID: _____ **Name:** _____

Instructions

1. This is a list of practice questions provided to help with your studying. The style, length, and difficulty of these problems may not necessarily reflect those of the actual midterm.
2. All questions are multiple-choice. Mark your answers in the separate answer sheet. **On the exam, we will grade solely based on your answer choices on the *separate* answer sheet.**

CS 185/285 Midterm (Practice Questions – Answer Sheet)

Student ID: _____ **Name:** _____

Only answers on the answer sheet will be graded. Fill in the bubble(s) for each problem. For single-choice questions, fill exactly one bubble. For "select all" questions, fill all bubbles corresponding to the correct letters. **Bubbles must be filled completely and unambiguously.** Your score will be deducted if the course staff and/or OCR misinterpret your answer due to ambiguous marking.

#	Problem	Circle answer(s)
1.1	Distribution Shift for Success/Failure Tasks (1)	a b c d e f
1.2	Distribution Shift for Success/Failure Tasks (2)	a b c d
1.3	Distribution Shift for Success/Failure Tasks (3)	a b c d e f
2.1	Open Loop Control (1)	a b c d
2.2	Open Loop Control (2)	a b c d e
3	Entropy regularized policy gradient	a b c d
4	Learning in Simulation	a b c d e f g
5.1	Computing the Policy Gradient (1)	a b c d e f
5.2	Computing the Policy Gradient (2)	a b c d e f
5.3	Computing the Policy Gradient (3)	a b c d e f g
6.1	Imitation learning analysis (1)	a b c d
6.2	Imitation learning analysis (2)	a b c d
7.1	IQL variants (1)	a b c d
7.2	IQL variants (2)	a b c d
8	LLM MDPs	a b c d e f g h i j k l m
9.1	Multi-Step Dynamics Models (1)	= / ≤ / ≥ / ? **and** = / ≤ / ≥ / ?
9.2	Multi-Step Dynamics Models (2)	= / ≤ / ≥ / ? **and** = / ≤ / ≥ / ?
10	Managing Distribution Shift in Offline RL	a b c d

If you have any additional comments (e.g., additional assumptions or interpretations you made, notes to the course staff, etc.), use the space below.

Potentially Useful Formulas

Some potentially useful formulas are provided below. These formulas are provided purely for convenience and may be ignored. They are **not** guaranteed to be directly applicable to individual problems, and you are fully responsible for determining their validity in each context or adapting them accordingly. In case of any conflict, you must follow the definitions and assumptions stated in each problem.

- Policy gradients

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau\sim p_\theta(\tau)}\left[\left(\sum_{t=1}^{H}\nabla_\theta \log \pi_\theta(\mathbf{a}_t \mid \mathbf{s}_t)\right)\left(\sum_{t=1}^{H} r(\mathbf{s}_t,\mathbf{a}_t)\right)\right]$$

$$\nabla_\theta J(\theta) \approx \mathbb{E}_{\tau\sim p_\theta(\tau)}\left[\sum_{t=1}^{H}\nabla_\theta \log \pi_\theta(\mathbf{a}_t \mid \mathbf{s}_t)\left(r(\mathbf{s}_t,\mathbf{a}_t) + \gamma \hat{V}^\pi(\mathbf{s}_{t+1}) - \hat{V}^\pi(\mathbf{s}_t)\right)\right]$$

- Q-learning

$$L(Q) = \mathbb{E}_{(\mathbf{s},\mathbf{a},\mathbf{s}')\sim\mathcal{D}}\left[\left(Q_\theta(\mathbf{s},\mathbf{a}) - [r(\mathbf{s},\mathbf{a}) + \gamma \max_{\mathbf{a}'} Q_{\bar{\theta}}(\mathbf{s}',\mathbf{a}')]\right)^2\right]$$

- Performance difference formula for off-policy policy gradients

$$J(\theta') - J(\theta) = \sum_{t=1}^{\infty}\gamma^{t-1}\mathbb{E}_{\mathbf{s}_t\sim p_{\theta'}(\mathbf{s}_t)}\left[\mathbb{E}_{\mathbf{a}_t\sim\pi_\theta(\mathbf{a}_t|\mathbf{s}_t)}\left[\frac{\pi_{\theta'}(\mathbf{a}_t \mid \mathbf{s}_t)}{\pi_\theta(\mathbf{a}_t \mid \mathbf{s}_t)}A^{\pi_\theta}(\mathbf{s}_t,\mathbf{a}_t)\right]\right]$$

- KL and TV divergences

$$D_{\mathrm{KL}}(p(x)\|q(x)) = \mathbb{E}_{x\sim p(x)}[\log p(x) - \log q(x)]$$

$$D_{\mathrm{TV}}(p(x),q(x)) = \frac{1}{2}\sum_x |p(x) - q(x)|$$

- Variational inference

$$\log p(x) \geq \mathbb{E}_{z\sim q_\phi(z|x)}[\log p_\theta(x \mid z) + \log p(z)] + \mathcal{H}(q_\phi(z \mid x)).$$

- Inverse RL

$$\nabla_\psi L \approx \frac{1}{N}\sum_{i=1}^{N}\nabla_\psi r_\psi(\tau_i) - \frac{1}{\sum_j w_j}\sum_{j=1}^{M} w_j \nabla_\psi r_\psi(\tau_j), \text{ where } w_j = \frac{p(\tau)\exp(r_\psi(\tau_j))}{\pi(\tau_j)}$$

1 Distribution Shift for Success/Failure Tasks

Imagine that instead of maximizing reward, we want to train policies that "succeed" according to a **binary** success criterion. Success can be evaluated on the last time step of a trajectory $\tau = (\mathbf{s}_1, \mathbf{a}_1, \ldots, \mathbf{s}_H, \mathbf{a}_H)$: $\mathbf{s}_H \in \Omega$ is successful for some set Ω. We will write $p_\theta(\tau \text{ successful})$ to denote the probability that the last state in a trajectory sampled by running $\pi_\theta(\mathbf{a}|\mathbf{s})$ is contained in Ω.

π_θ will be trained with **supervised learning** using a dataset $\mathcal{D} = \{(\mathbf{s}_i, \mathbf{a}_i)\}_{i=1}^N$ generated by a deterministic optimal Markovian policy $\pi^\star(\mathbf{s}_i)$ that always succeeds, such that $\mathbf{a}_i = \pi^\star(\mathbf{s}_i)$. Use $p_{\text{train}}(\mathbf{s}_t)$ to denote the distribution of training states, and use $p_{\pi_\theta}(\mathbf{s}_t)$ to denote the distribution of states under the policy π_θ at time step t. Assume $\mathbb{E}_{\mathbf{s}_t \sim p_{\text{train}}(\mathbf{s}_t)}[\pi_\theta(\mathbf{a}_t \neq \pi^\star(\mathbf{s}_t)|\mathbf{s}_t)] \leq \epsilon$ for all t.

We'll derive an upper bound on $p_\theta(\tau \text{ failed})$, based on the behavioral cloning cost bound derivation.

(1) First, letting $\mathbb{1}(\mathbf{s}_H \notin \Omega)$ be 1 if $\mathbf{s}_H$ is not in Ω (0 otherwise), and $\mathbb{1}(\mathbf{s}_H \in \Omega)$ be 1 if $\mathbf{s}_H$ is in Ω (0 otherwise), which of the following quantities is equal to the probability that the policy fails?

(a) $p_\theta(\tau \text{ failed}) = p_{\text{train}}(\mathbf{s}_H)\mathbb{1}(\mathbf{s}_H \notin \Omega)$

(b) $p_\theta(\tau \text{ failed}) = \sum_{\mathbf{s}_H} p_{\text{train}}(\mathbf{s}_H)\mathbb{1}(\mathbf{s}_H \notin \Omega)$

(c) $p_\theta(\tau \text{ failed}) = \sum_{\mathbf{s}_H} p_{\text{train}}(\mathbf{s}_H)\mathbb{1}(\mathbf{s}_H \in \Omega)$

(d) $p_\theta(\tau \text{ failed}) = p_{\pi_\theta}(\mathbf{s}_H)\mathbb{1}(\mathbf{s}_H \notin \Omega)$

(e) $p_\theta(\tau \text{ failed}) = \sum_{\mathbf{s}_H} p_{\pi_\theta}(\mathbf{s}_H)\mathbb{1}(\mathbf{s}_H \notin \Omega)$

(f) $p_\theta(\tau \text{ failed}) = \sum_{\mathbf{s}_H} p_{\pi_\theta}(\mathbf{s}_H)\mathbb{1}(\mathbf{s}_H \in \Omega)$

(2) If we follow similar logic as we saw in the behavioral cloning error bound proof in lecture to derive an upper bound on $p_\theta(\tau \text{ failed})$, using the fact that $\sum_{\mathbf{s}_t} |p_{\pi_\theta}(\mathbf{s}_t) - p_{\text{train}}(\mathbf{s}_t)| \leq \epsilon t$ as derived in lecture, which of the following is a step in the derivation of the bound?

(a) $p_\theta(\tau \text{ failed}) \leq \sum_{\mathbf{s}_H} \big(|p_{\pi_\theta}(\mathbf{s}_H) - p_{\text{train}}(\mathbf{s}_H)| + p_{\text{train}}(\mathbf{s}_H)\big)\mathbb{1}(\mathbf{s}_H \in \Omega)$

(b) $p_\theta(\tau \text{ failed}) \leq \sum_{\mathbf{s}_H} \big(|p_{\pi_\theta}(\mathbf{s}_H) - p_{\text{train}}(\mathbf{s}_H)| + p_{\pi_\theta}(\mathbf{s}_H)\big)\mathbb{1}(\mathbf{s}_H \in \Omega)$

(c) $p_\theta(\tau \text{ failed}) \leq \sum_{\mathbf{s}_H} \big(|p_{\pi_\theta}(\mathbf{s}_H) - p_{\text{train}}(\mathbf{s}_H)| + p_{\text{train}}(\mathbf{s}_H)\big)\mathbb{1}(\mathbf{s}_H \notin \Omega)$

(d) $p_\theta(\tau \text{ failed}) \leq \sum_{\mathbf{s}_H} \big(|p_{\pi_\theta}(\mathbf{s}_H) - p_{\text{train}}(\mathbf{s}_H)| + p_{\pi_\theta}(\mathbf{s}_H)\big)\mathbb{1}(\mathbf{s}_H \notin \Omega)$

(3) Based on the above, which of these is the best (i.e., tightest among others) upper bound on $p_\theta(\tau \text{ failed})$?

(a) $p_\theta(\tau \text{ failed}) \leq \epsilon$

(b) $p_\theta(\tau \text{ failed}) \leq \epsilon H$

(c) $p_\theta(\tau \text{ failed}) \leq \epsilon H^2$

(d) $p_\theta(\tau \text{ failed}) \leq 1 - \epsilon$

(e) $p_\theta(\tau \text{ failed}) \leq 1 - \epsilon H$

(f) $p_\theta(\tau \text{ failed}) \leq 1 - \epsilon H^2$

2 Open Loop Control

Consider an MDP $(\mathcal{S}, \mathcal{A}, r, \mathrm{P}, \gamma)$, where $r : \mathcal{S} \to \mathbb{R}$ and $0 < \gamma < 1$. Recall that a sequence of actions $\mathbf{A}_t$ defines an *open loop policy.* Each distribution $\pi(a \mid s)$ conditioned on state defines a *closed loop policy.*

As usual, the *discounted total return* can be defined equivalently for both types of policies,

$$J(\pi) = \sum_{t=0}^{\infty} \gamma^t r(\mathbf{s}_t) \quad \text{and} \quad J(\mathbf{A}^*) = \sum_{t=0}^{\infty} \gamma^t r(\mathbf{s}_t)$$

as well as the *discounted state occupancy* measure:

$$p^{\pi}(s) = \sum_{t=0}^{\infty} \gamma^t p(\mathbf{s}_t = s) \quad \text{and} \quad p^{\mathbf{A}}(s) = \sum_{t=0}^{\infty} \gamma^t p(\mathbf{s}_t = s)$$

(1) Which of the following is *not* true? Select one response.

(a) There may be no stationary (time-invariant) optimal open loop policies in an MDP.

(b) In an MDP with deterministic dynamics and initial state, the best open loop policy attains the same performance as the best closed loop policy, i.e., $J(\pi^*) = J(\mathbf{A}^*)$

(c) There exists an MDP with nonnegative, bounded rewards where the best closed-loop policy gets return 1 but the best open loop policy gets return 0, i.e., $J(\pi^*) = 1$ and $J(\mathbf{A}^*) = 0$

(d) Consider an open loop policy $\mathbf{A}$. There always exists a deterministic, Markovian closed-loop policy π that attains the same discounted (marginal) state occupancy, so $p^{\mathbf{A}} = p^{\pi}$

(2) The cross entropy method when applied to open loop control _____. Select one response. You may assume a max horizon of H with $1 < M < H$ with M elites selected at each iteration.

(a) With unlimited iterations, will almost surely (with probability arbitrarily close to 1) converge to an optimal sequence of actions when the sampling distribution $p(\mathbf{A}_t)$ is a multivariate Gaussian $\mathcal{N}(\mu, \Sigma)$ and sample mean μ and covariance Σ are computed from the previous iteration's elites.

(b) Is the same as random shooting when the sampling distribution $p(\mathbf{A}_t)$ is uniform over $\mathcal{A}$

(c) Will only converge when the sampling distribution depends on the previous iteration's elites.

(d) Doesn't require gradients through the model

(e) Can only be used with discrete action spaces

3 Entropy regularized policy gradient

Recall that $\mathcal{H}(\pi_\theta(\mathbf{a}|\mathbf{s})) = -\mathbb{E}_{\mathbf{a}\sim\pi_\theta(\mathbf{a}|\mathbf{s})}[\log \pi_\theta(\mathbf{a}|\mathbf{s})]$. Assuming that the action space is intractably large and discrete (so no closed form expression for the entropy) and there is one time step (i.e., a bandit, such that $Q(\mathbf{s},\mathbf{a}) = r(\mathbf{s},\mathbf{a})$), derive a tractable unbiased policy gradient expression with samples for the entropy-regularized objective of the form

$$J(\theta) = \mathbb{E}_{\mathbf{s}\sim p(\mathbf{s})}\left[\mathbb{E}_{\mathbf{a}\sim\pi_\theta(\mathbf{a}|\mathbf{s})}[r(\mathbf{s},\mathbf{a})] + \mathcal{H}(\pi_\theta(\mathbf{a}|\mathbf{s}))\right].$$

Use a constant baseline b.

(a) $\nabla_\theta J(\theta) \approx \frac{1}{N}\sum_{i=1}^{N} \nabla_\theta \log \pi_\theta(\mathbf{a}|\mathbf{s})[r(\mathbf{s},\mathbf{a}) + \log \pi_\theta(\mathbf{a}|\mathbf{s}) - b]$

(b) $\nabla_\theta J(\theta) \approx \frac{1}{N}\sum_{i=1}^{N} \nabla_\theta \log \pi_\theta(\mathbf{a}|\mathbf{s})[r(\mathbf{s},\mathbf{a}) - \log \pi_\theta(\mathbf{a}|\mathbf{s}) - b]$

(c) $\nabla_\theta J(\theta) \approx \frac{1}{N}\sum_{i=1}^{N} \nabla_\theta \log \pi_\theta(\mathbf{a}|\mathbf{s})[r(\mathbf{s},\mathbf{a}) - b]$

(d) $\nabla_\theta J(\theta) \approx \frac{1}{N}\sum_{i=1}^{N} \nabla_\theta \log \pi_\theta(\mathbf{a}|\mathbf{s})[-\log \pi_\theta(\mathbf{a}|\mathbf{s}) - b]$

4 Learning in Simulation

Suppose we have an MDP $M = (\mathcal{S}, \mathcal{A}, r, \mathrm{P}, \gamma)$ with $0 \le r(s) \le 1$. We consider the problem of learning in a model $f(s' \mid s, a)$ of the dynamics $P(s' \mid s, a)$. Learning with a model f can be viewed as learning in the modified MDP $F = (\mathcal{S}, \mathcal{A}, r, f, \gamma)$.

Denote by π_T^* the optimal policy in an MDP T. The *regret* of any policy π in T is the amount by which it is suboptimal relative to π_T^*, stated as

$$\tilde{R}^T(\pi) = J^T(\pi_T^*) - J^T(\pi),$$

where

$$J^T(\pi) = \sum_{t=0}^{\infty} \gamma^t r(s_t)$$

is the standard discounted return for an MDP T.

When we train in the simulator with model f of the dynamics, we are learning π_F^*. Our hope with a good enough simulator is that it will transfer to the environment M, or in other words $\tilde{R}^M(\pi_F^*)$ will be low.

Suppose $D_{KL}(\mathrm{P}(s'|s,a)\|f(s' \mid s,a)) < \varepsilon^2$ for any s, a . Which of the following upper bounds on the regret $\tilde{R}^M(\pi_F^*)$ hold? Select all that apply.

Hint: Pinsker's inequality gives

$$2\delta(p,q) = \sum_{\mathcal{X}} |p(x) - q(x)| \le \sqrt{2D_{KL}(p\|q)}$$

(a) $\tilde{R}^M(\pi_F^*) \le \varepsilon\gamma\sqrt{2}(1-\gamma)^{-2}$

(b) $\tilde{R}^M(\pi_F^*) \le \varepsilon\gamma\sqrt{1/2}(1-\gamma)^{-2}$

(c) $\tilde{R}^M(\pi_F^*) \le \varepsilon\sqrt{1/2}(1-\gamma)^{-2}$

(d) $\tilde{R}^M(\pi_F^*) \le 2\varepsilon\gamma(1-\gamma)^{-1}$

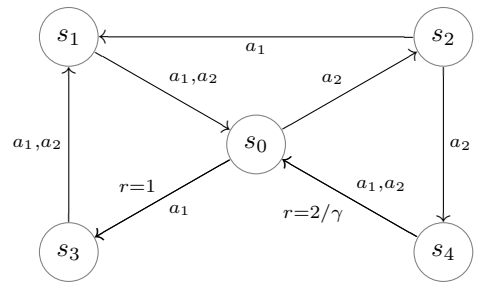
(e) $\tilde{R}^M(\pi_F^*) \le \varepsilon\gamma\sqrt{1/2}(1-\gamma)^{-3/2}$

(f) $\tilde{R}^M(\pi_F^*) \le \varepsilon\gamma\sqrt{2}(1-\gamma)^{-1}$

(g) $\tilde{R}^M(\pi_F^*) \le \varepsilon\sqrt{2}(1-\gamma)^{-5/2}$

5 Computing the Policy Gradient

Consider the following MDP:



We assume that the initial state is s_0, and there are actions $A = \{a_1, a_2\}$ that correspond to the edges marked on the diagram. Assume the rewards are zero except for the edges annotated on the diagram.

We learn a policy π_θ parameterized by $\theta \in [0, 1]$ such that at any state $s \in \{s_0, s_1, s_2, s_3, s_4\}$:

$$\pi_\theta(a_1 \mid s) = 1 - \theta$$
$$\pi_\theta(a_2 \mid s) = \theta.$$

We wish to optimize the discounted return:

$$J(\theta) = \mathbb{E}\Big[\sum_{t=0}^{\infty} \gamma^t r(\mathbf{s}_t, \mathbf{a}_t)\Big].$$

(1) Compute the policy gradient $\nabla_\theta J(\theta)$ for this MDP.

(a) $\dfrac{\theta(1-2\theta)-1}{1-\gamma^3}$

(b) $\dfrac{2\theta-1}{1-\gamma^3}$

(c) $\dfrac{1+\theta(\gamma-2\theta)}{1-\gamma^2}$

(d) $\dfrac{4\theta\gamma-1}{1-\gamma^3}$

(e) $\dfrac{\theta(2\theta-1)}{1-\gamma^2}$

(f) $\dfrac{2\theta\gamma-1}{1-\gamma^2}$

(2) What is the maximum achievable return $J(\theta^*)$? Assume $\gamma > 1/2$.

(a) $\dfrac{1}{1-\gamma}$

(b) $\dfrac{2\gamma}{1-\gamma}$

(c) $\dfrac{1}{1-\gamma^2}$

(d) $\dfrac{2}{1-\gamma^2}$

(e) $\dfrac{\gamma}{1-\gamma^3}$

(f) $\dfrac{2\gamma}{1-\gamma^3}$

(3) For which initial values of θ does the policy gradient recover the optimal policy when $\gamma = 3/4$? Assume updates are performed with gradient descent $\theta \leftarrow \theta + \alpha\nabla J(\theta)$ for a sufficiently small learning rate α.

(a) $\theta > 1/4$

(b) $\theta < 2/3$

(c) $\theta > 1/3$

(d) $\theta < 1/2$

(e) $\theta < 1/3$

(f) $\theta > 3/4$

(g) $\theta < 3/4$

6 Imitation learning analysis

Consider the problem of imitation learning within a discrete MDP with horizon T and an expert policy π^*. We gather expert demonstrations from π^* and fit an imitation policy π_θ to these trajectories so that

$$\mathbb{E}_{p_{\pi^*}(s)}\, \pi_\theta(\mathbf{a} \neq \pi^*(s) \mid s) = \frac{1}{T}\sum_{t=1}^{T} \mathbb{E}_{p_{\pi^*}(s_t)}\, \pi_\theta(\mathbf{a}_t \neq \pi^*(s_t) \mid s_t) \leq \varepsilon,$$

i.e., the expected likelihood that the learned policy π_θ disagrees with the expert π^* within the training distribution p_{π^*} of states drawn from random expert trajectories is at most ε.

For convenience, the notation $p_\pi(s_t)$ indicates the state distribution under π at time step t while $p(s)$ indicates the state marginal of π across time steps, unless indicated otherwise.

Consider the expected return of the learned policy π_θ for a state-dependent reward $r(s_t)$, where we assume the reward is bounded with $|r(s_t)| \leq 1$:

$$J(\pi) = \sum_{t=1}^{T} \mathbb{E}_{p_\pi(s_t)} r(s_t).$$

(1) Express the upper bound of $J(\pi^*) - J(\pi_\theta)$ in terms of T and ε. It is enough to express the upper bounds using big O notation (in T and ε).

(a) $\mathcal{O}(T^2\varepsilon)$

(b) $\mathcal{O}(T\varepsilon)$

(c) $\mathcal{O}(T^3\varepsilon)$

(d) $\mathcal{O}(T^2\varepsilon^2)$

(2) If the reward only depends on the last state (i.e., $r(\mathbf{s}_t) = 0$ for all $t < T$), how does the upper bound change (or not)? Select the tightest new bound.

(a) $\mathcal{O}(T^2\varepsilon)$

(b) $\mathcal{O}(T\varepsilon)$

(c) $\mathcal{O}(T^3\varepsilon)$

(d) $\mathcal{O}(T^2\varepsilon^2)$

7 IQL variants

Implicit Q-learning (IQL) is an offline RL algorithm that trains a Q-function $Q_\phi(\mathbf{s}, \mathbf{a})$ and a value function $V_\psi(\mathbf{s})$ using expectile regression. Consider the following four variants of IQL losses:

(a)
$$\mathcal{L}(\phi) = \mathbb{E}_{\mathbf{s},\mathbf{a},\mathbf{s}'\sim\mathcal{D}}[(Q_\phi(\mathbf{s}, \mathbf{a}) - r(\mathbf{s}, \mathbf{a}) - \gamma V_\psi(\mathbf{s}'))^2]$$
$$\mathcal{L}(\psi) = \mathbb{E}_{\mathbf{s},\mathbf{a}\sim\mathcal{D}}[\ell_2^\tau(V_\psi(\mathbf{s}) - Q_{\bar{\phi}}(\mathbf{s}, \mathbf{a}))]$$

(b)
$$\mathcal{L}(\phi) = \mathbb{E}_{\mathbf{s},\mathbf{a},\mathbf{s}'\sim\mathcal{D}}[(Q_\phi(\mathbf{s}, \mathbf{a}) - r(\mathbf{s}, \mathbf{a}) - \gamma V_\psi(\mathbf{s}'))^2]$$
$$\mathcal{L}(\psi) = \mathbb{E}_{\mathbf{s},\mathbf{a}\sim\mathcal{D}}[\ell_2^\tau(Q_{\bar{\phi}}(\mathbf{s}, \mathbf{a}) - V_\psi(\mathbf{s}))]$$

(c)
$$\mathcal{L}(\phi) = \mathbb{E}_{\mathbf{s},\mathbf{a},\mathbf{s}',\mathbf{a}'\sim\mathcal{D}}[\ell_2^\tau(Q_\phi(\mathbf{s}, \mathbf{a}) - r(\mathbf{s}, \mathbf{a}) - \gamma Q_{\bar{\phi}}(\mathbf{s}', \mathbf{a}'))].$$

(d)
$$\mathcal{L}(\phi) = \mathbb{E}_{\mathbf{s},\mathbf{a},\mathbf{s}',\mathbf{a}'\sim\mathcal{D}}[\ell_2^\tau(r(\mathbf{s}, \mathbf{a}) + \gamma Q_{\bar{\phi}}(\mathbf{s}', \mathbf{a}') - Q_\phi(\mathbf{s}, \mathbf{a}))].$$

In the above losses, $\mathcal{D}$ denotes an offline dataset, $(\mathbf{s}, \mathbf{a}, \mathbf{s}', \mathbf{a}')$ denotes an adjacent (state, action, next state, next action) pair sampled from the dataset, $\ell_2^\tau(x) = |\mathbb{1}(x > 0) - \tau|x^2$ denotes the expectile loss with expectile $\tau \in [0.5, 1)$, and $Q_{\bar{\phi}}(\mathbf{s}, \mathbf{a})$ denotes the target Q-function.

(1) Among the four variants, which ones approximate the optimal Q-function in an unbiased manner when $\tau \to 1$ (under the assumption that $\mathcal{D}$ has sufficient coverage)? Select **all** of the variants (a-d) that apply. (Hint: Think about whether they are unbiased in stochastic environments.)

(2) Notice that when $\tau = 0.5$, IQL learns the behavioral Q-function $Q^{\pi_\beta}(\mathbf{s}, \mathbf{a})$, where π_β is the data-collecting policy for the dataset $\mathcal{D}$. Among the four variants, which ones approximate the behavioral Q-function in an unbiased manner when $\tau = 0.5$ (under the assumption that $\mathcal{D}$ has sufficient coverage)? Select **all** of the variants (a-d) that apply.

8 LLM MDPs

Consider an LLM MDP $\mathcal{M}_{\text{LLM}} = (\mathcal{S}, \mathcal{A}, \mathcal{T}, r)$ defined as follows.

The state space

$$\mathcal{S} = \{(x_1, x_2, \ldots, x_n) : x_i \in \mathcal{W}, 1 \leq n \leq H\}$$

is the set of all possible token sequences, where $\mathcal{W}$ is the token vocabulary and H is the maximum sequence length.

The action space is

$$\mathcal{A} = \mathcal{W},$$

i.e., actions correspond to selecting the next token.

The transition dynamics $\mathcal{T}$ is deterministic concatenation: for a state $\mathbf{s} = (x_1, x_2, \ldots, x_n)$ and an action $\mathbf{a} = w$, the next state is deterministically given by $\mathbf{s}' = (x_1, x_2, \ldots, x_n, w)$. The initial state is always a fixed prompt. For simplicity, assume that there is no end-of-sequence token, so all final token sequences have length exactly H.

The reward r is 1 only when the current token sequence is of full length and correctly answers the task, and 0 otherwise.

For this LLM MDP $\mathcal{M}_{\text{LLM}}$, select all correct statements:

(a) Subtracting a baseline from a policy gradient estimator provides no benefit.

(b) Learning a dynamics model provides no benefit.

(c) Training a value function is impossible in practice.

(d) When we have a state-value function ($V(\mathbf{s})$), additionally training an action-value function that outputs a scalar value ($Q(\mathbf{s}, \mathbf{a})$) provides no benefit.

(e) There is no bias-variance tradeoff in policy gradient estimators.

(f) We do not need *variational* inference in the control-as-inference framework.

(g) DQN is incompatible with the LLM MDP.

(h) TRPO is incompatible with the LLM MDP.

(i) Offline RL is incompatible with the LLM MDP.

(j) Next token prediction is equivalent to behavioral cloning.

(k) Value iteration becomes equivalent to policy iteration.

(l) Policy iteration can always find the optimal policy in one policy update.

(m) Maximizing the entropy of the policy's *action* distribution ($\mathcal{H}(\pi(\mathbf{a}|\mathbf{s}))$) at each state $\mathbf{s}$ is always enough to maximize the entropy of the final *state* distribution ($\mathcal{H}(p^\pi(\mathbf{s}_H))$).

9 Multi-Step Dynamics Models

Let $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{T}, r)$ be a finite-horizon MDP with horizon H. For each $n \in \{1, 2, 3\}$, define the length-n action-chunked MDP $\mathcal{M}^{(n)} = (\mathcal{S}, \mathcal{A}^n, \mathcal{T}^{(n)}, r^{(n)})$ as follows.

The action space is the set of length-n action chunks,

$$\mathcal{A}^n = \{(\mathbf{a}_0, \ldots, \mathbf{a}_{n-1}) : \mathbf{a}_i \in \mathcal{A}\}.$$

The transition dynamics $\mathcal{T}^{(n)}$ is the marginalized n-step transition dynamics induced by $\mathcal{M}$:

$$\mathcal{T}^{(n)}(\mathbf{s}_{t+n}|\mathbf{s}_t, (\mathbf{a}_t, \ldots, \mathbf{a}_{t+n-1})) = \int \left(\prod_{i=0}^{n-1} \mathcal{T}(\mathbf{s}_{t+i+1}|\mathbf{s}_{t+i}, \mathbf{a}_{t+i}) \right) d\mathbf{s}_{t+1} d\mathbf{s}_{t+2} \cdots d\mathbf{s}_{t+n-1}.$$

For simplicity, assume that H is divisible by all $n \in \{1, 2, 3\}$.

The reward function $r^{(n)}$ is the expected sum of rewards while executing the chunk:

$$r^{(n)}(\mathbf{s}_t, (\mathbf{a}_t, \ldots, \mathbf{a}_{t+n-1})) = \int \left(\prod_{i=0}^{n-2} \mathcal{T}(\mathbf{s}_{t+i+1}|\mathbf{s}_{t+i}, \mathbf{a}_{t+i}) \right) \left(\sum_{i=0}^{n-1} r(\mathbf{s}_{t+i}, \mathbf{a}_{t+i}) \right) d\mathbf{s}_{t+1} d\mathbf{s}_{t+2} \cdots d\mathbf{s}_{t+n-1}.$$

Let $J^{(n)}(\pi_{(n)})$ denote the expected return of an action-chunking policy $\pi_{(n)}((\mathbf{a}_t, \mathbf{a}_{t+1}, \ldots, \mathbf{a}_{t+n-1})|\mathbf{s}_t)$ in $\mathcal{M}^{(n)}$. That is,

$$J^{(n)}(\pi_{(n)}) = \mathbb{E}_{\tau^{(n)} \sim p_{\pi_{(n)}}(\tau^{(n)})} \left[\sum_{t=0}^{\frac{H}{n}-1} r^{(n)}(\mathbf{s}_t^{(n)}, \mathbf{a}_t^{(n)}) \right],$$

where

$$\tau^{(n)} = (\underbrace{\mathbf{s}_0}_{\mathbf{s}_0^{(n)}}, \underbrace{(\mathbf{a}_0, \mathbf{a}_1, \ldots, \mathbf{a}_{n-1})}_{\mathbf{a}_0^{(n)}}, \underbrace{\mathbf{s}_n}_{\mathbf{s}_1^{(n)}}, \underbrace{(\mathbf{a}_n, \mathbf{a}_{n+1}, \ldots, \mathbf{a}_{2n-1})}_{\mathbf{a}_1^{(n)}}, \ldots, \underbrace{\mathbf{s}_{H-n}}_{\mathbf{s}_{\frac{H}{n}-1}^{(n)}}, \underbrace{(\mathbf{a}_{H-n}, \ldots, \mathbf{a}_{H-1})}_{\mathbf{a}_{\frac{H}{n}-1}^{(n)}})$$

is sampled from the action-chunking policy $\pi_{(n)}$ in the action-chunked MDP $\mathcal{M}^{(n)}$. Let $\pi^*_{(n)}$ be an optimal action-chunking policy that maximizes $J^{(n)}$ in $\mathcal{M}^{(n)}$.

Fill in each box below by choosing the most suitable symbol from $\{=, \leq, \geq, ?\}$. Choose ? when none of the relations $(=, \leq, \geq)$ always hold between the two terms. Do **not** assume that $\pi^*_{(n)}$ is necessarily a deterministic policy (recall that an optimal policy need not be unique, and some optimal policies may be stochastic when multiple actions lead to the same value, even in deterministic environments).

(1) For any $\mathcal{M}$,

$$J^{(1)}(\pi^*_{(1)}) \;\boxed{}\; J^{(2)}(\pi^*_{(2)}) \;\boxed{}\; J^{(3)}(\pi^*_{(3)}).$$

(2) If the dynamics $\mathcal{T}$ is deterministic,

$$J^{(1)}(\pi^*_{(1)}) \;\boxed{}\; J^{(2)}(\pi^*_{(2)}) \;\boxed{}\; J^{(3)}(\pi^*_{(3)}).$$

10 Managing Distribution Shift in Offline RL

In this question, we will analyze how reward bonuses can be used to manage distribution shift in offline RL. We consider an MDP $M = (\mathcal{S}, \mathcal{A}, r, p)$ with offline data $\mathcal{D}$ collected by a policy π_β. We denote by p_π the distribution over states induced by π.

As a motivating example, consider the soft actor-critic (SAC) algorithm discussed in HW 3. When updating, SAC adds an adjustment of $b(\mathbf{s}, \mathbf{a}) = -\log \pi(\mathbf{a} \mid \mathbf{s})$ to the target values $\mathbb{E}_{\pi(\mathbf{a}|\mathbf{s})}[Q(\mathbf{s}, \mathbf{a}) + b(\mathbf{s}, \mathbf{a})]$ to enforce a maximum entropy regularizer on the policy. Alternatively, we could impose a similar form of entropy regularization by adding the bonus directly to the reward. In expectation, we would optimize

$$\begin{aligned}\mathbb{E}_{s,a\sim p_\pi}[r(\mathbf{s}, \mathbf{a}) + \lambda b(\mathbf{s}, \mathbf{a})] &= \tfrac{1}{H} J(\pi) - \lambda\, \mathbb{E}_{s\sim p_\pi} \mathbb{E}_{\pi(\mathbf{a}|\mathbf{s})} \log \pi(\mathbf{a} \mid \mathbf{s}) \\ &= \tfrac{1}{H} J(\pi) + \lambda\, \mathbb{E}_{s\sim p_\pi} \mathcal{H}[\pi(\mathbf{a} \mid \mathbf{s})].\end{aligned}$$

Thus, entropy regularization can also be implemented by adding the bonus $b(\mathbf{s}, \mathbf{a})$ to rewards. In the following parts, you will show how a similar reward bonus can be used to constrain distribution shift in offline RL.

We wish to learn a Q function and policy π from the offline data $\mathcal{D}$ under some constraint $D(\pi, \pi_\beta) \leq \varepsilon$ with the following update:

$$Q(\mathbf{s}, \mathbf{a}) \leftarrow r(\mathbf{s}, \mathbf{a}) + \mathbb{E}_{\mathbf{a}'\sim\pi}\left[Q\left(\mathbf{s}', \mathbf{a}'\right)\right] \tag{1}$$

$$\text{where} \quad \pi = \arg\max_\pi \mathbb{E}_{\mathbf{s},\mathbf{a}\sim p_\pi}[Q(\mathbf{s}, \mathbf{a})] \;\; \text{s.t.} \;\; D(\pi, \pi_\beta) \leq \varepsilon. \tag{2}$$

Directly enforcing the constraint in (2) is challenging with the environment rewards $r(\mathbf{s}, \mathbf{a})$, so we will implicitly enforce the constraint with a Lagrangian, modifying the reward to $\bar{r}(\mathbf{s}, \mathbf{a}) = r(\mathbf{s}, \mathbf{a}) + \lambda b(\mathbf{s}, \mathbf{a})$ in (1). The overall optimization then becomes:

$$Q(\mathbf{s}, \mathbf{a}) \leftarrow \bar{r}(\mathbf{s}, \mathbf{a}) + \mathbb{E}_{\mathbf{a}'\sim\pi}\left[Q\left(\mathbf{s}', \mathbf{a}'\right)\right] \tag{3}$$

$$\text{where} \quad \pi = \arg\max_\pi \mathbb{E}_{\mathbf{s},\mathbf{a}\sim p_\pi}[Q(\mathbf{s}, \mathbf{a})]. \tag{4}$$

You may assume that $\lambda > 0$ is selected appropriately to enforce the constraint as follows:

$$\left(\arg\max_\pi \mathbb{E}_{\mathbf{s},\mathbf{a}\sim p_\pi}[Q(\mathbf{s}, \mathbf{a})] - \lambda D(\pi, \pi_\beta)\right) = \left(\arg\max_\pi \mathbb{E}_{\mathbf{s},\mathbf{a}\sim p_\pi}[Q(\mathbf{s}, \mathbf{a})] \;\; \text{s.t.} \;\; D(\pi, \pi_\beta) \leq \varepsilon\right).$$

You may also assume access to the distributions $\pi(\mathbf{a} \mid \mathbf{s})$ and $\pi_\beta(\mathbf{a} \mid \mathbf{s})$ in your answers.

Suppose we wish to learn π under the following KL-divergence constraint:

$$D(\pi, \pi_\beta) = \mathbb{E}_{\mathbf{s}\sim p_\pi} D_{KL}[\pi(\mathbf{a} \mid \mathbf{s}) \| \pi_\beta(\mathbf{a} \mid \mathbf{s})].$$

How can we enforce this constraint by adding a bonus $b(\mathbf{s}, \mathbf{a})$ to the reward $\bar{r}(\mathbf{s}, \mathbf{a}) = r(\mathbf{s}, \mathbf{a}) + \lambda b(\mathbf{s}, \mathbf{a})$?

(a) $b(\mathbf{s}, \mathbf{a}) = -\log \pi(\mathbf{a} \mid \mathbf{s})$

(b) $b(\mathbf{s}, \mathbf{a}) = \log \pi_\beta(\mathbf{a} \mid \mathbf{s})$

(c) $b(\mathbf{s}, \mathbf{a}) = \log \pi(\mathbf{a} \mid \mathbf{s}) - \log \pi_\beta(\mathbf{a} \mid \mathbf{s})$

(d) $b(\mathbf{s}, \mathbf{a}) = \log \pi_\beta(\mathbf{a} \mid \mathbf{s}) - \log \pi(\mathbf{a} \mid \mathbf{s})$